

Non-invasive physiological assessment of intermediate coronary stenoses from plain angiography through artificial intelligence: the STARFLOW system

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Background

Despite evidence supporting use of fractional flow reserve (FFR) and instantaneous waves-free ratio (iFR) to improve outcome of patients undergoing coronary angiography (CA) and percutaneous coronary intervention, such techniques are still underused in clinical practice due to economic and logistic issues.

Objectives

We aimed to develop an artificial intelligence (AI)-based application to compute FFR and iFR from plain CA.

Methods and results

Consecutive patients performing FFR or iFR or both were enrolled. A specific multi-task deep network exploiting 2 projections of the coronary of interest from standard CA was appraised. Accuracy of prediction of FFR/iFR of the AI model was the primary endpoint, along with sensitivity and specificity. Prediction was tested both for continuous values and for dichotomous classification (positive/negative) for FFR or iFR. Subgroup analyses were performed for FFR and iFR.

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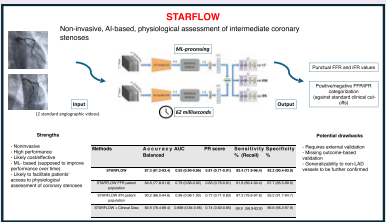
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A total of 389 patients from 5 centers were enrolled. Mean age was 67.9 ± 9.6 and 39.2% of patients were admitted for acute coronary syndrome. Overall, the accuracy was 87.3% (81.2–93.4%), with a sensitivity of 82.4% (71.9–96.4%) and a specificity of 92.2% (90.4–93.9%). For FFR, accuracy was 84.8% (77.8–91.8%), with a sensitivity of 81.9% (69.4–94.4%) and a specificity of 87.7% (85.5–89.9%), while for iFR accuracy was 90.2% (86.0–94.6%), with a sensitivity of 87.2% (76.6–97.8%) and a specificity of 93.2% (91.7–94.7%, all confidence intervals 95%).

Conclusion

The presented machine-learning based tool showed high accuracy in prediction of wire-based FFR and iFR.

Graphical Abstract



Keywords

Artificial intelligence • Fractional flow reserve • Instantaneous waves-free ratio • Percutaneous coronary intervention • Coronary physiology

Key Learning Points

What is already known

- Fractional flow reserve (FFR) and instantaneous waves-free ratio (iFR) are a standard of care for assessing the haemodynamic relevance of intermediate coronary stenoses and drive percutaneous coronary revascularization.
- These ancillary techniques are underused in clinical practice probably due to the procedural time, costs associated with probes and the procedural risk of the invasive assessment.

What this study adds

- The STARFLOW tool here presented is a machine-learning based application meant to estimate FFR and iFR from plain coronary angiography.
- The tool processes two angiographic views and provides FFR, iFR, and a binary classification (positive or negative) according to standard validated cutoffs.
- In this study, STARFLOW proved satisfying accuracy and it is expected to improve patients access to physiological assessment of coronary stenoses by reducing the time and costs required for estimation and limiting the risks associated with the invasive assessment.

Introduction

Invasive coronary assessment of fractional flow reserve (FFR) and/or instantaneous free wave ratio (iFR) should be pursued to guide percutaneous coronary revascularization (PCI) of intermediate grade coronary lesions.^{1,2} Despite the demonstration of a reduction in subsequent revascularization procedures and of a prognostic benefit obtained with PCI driven by FFR/iFR,^{3–8} the practical uptake of methodology has been modest.⁹ The main limiting factors appear to be the time for set-up and measurement and the non-negligible cost of the diagnostic probe.

Artificial intelligence (AI) and machine learning (ML) have shown the capability to significantly improve risk assessment and stratification as well as the analysis of cardiovascular imaging, in particular regarding quality of images, segmentation, and shape analysis.¹⁰

In the field of coronary physiology and pathophysiology assessment, the possibility to obtain a reliable estimate of FFR/iFR without the need of the whole setup and invasive measurements, appears particularly appealing. In this contest, promising results have been obtained, but they were mostly based on angio CT scans performed in highly selected patients,¹¹ or required large pre-processing (i.e.

reconstructions of coronary arteries), consequently with a limited use in everyday clinical practice.¹²

In the present paper, we tested the feasibility and the accuracy of an AI-based approach aimed to derive continuous and dichotomous values of FFR/iFR in an unselected cohort of patients from the regular coronary angiography (CA).

Methods

Selection of patients and data collection

Consecutive patients undergoing invasive physiological assessment for intermediate coronary stenoses (diameter of stenosis $\geq 30\%$ and $\leq 90\%$ at angiographic visual estimation) with iFR, FFR, or both from January 2020 to January 2022 across 5 centers were retrospectively included.

Demographic and baseline clinical features were extracted from pre-specified electronic databases. Both patients with chronic coronary syndrome (CCS) and acute coronary syndrome (ACS) were included. For patients admitted with ACS, physiological assessment was performed only for non-culprit coronary lesions.

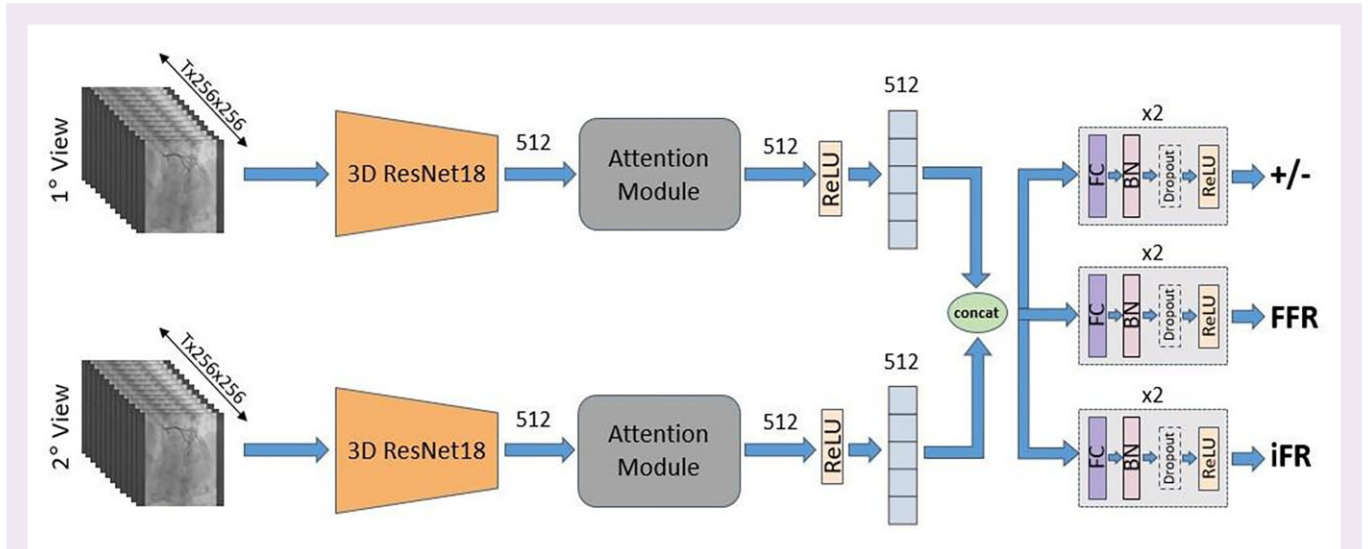


Figure 1 STARFLOW architecture. A (shared between the views) 3D CNN extracts spatio-temporal features, which are then focused through a space-time vision transformer. The obtained features are then fed to a three-output layer that, respectively, performs binary classification (over threshold) and iFR and FFR quantification.

Coronary angiography and physiological measurements were performed according to standardized clinical practice at all participating centers. All physiological measurements were made after diagnostic angiography. A coronary pressure wire (Philips Volcano®) was first advanced and equalized to the aortic pressure with the pressure sensor at the tip of the guide catheter. The wire was then advanced at least three vessel diameters distal to the stenosis and a recording of distal coronary and aortic pressure obtained. Instantaneous waves-free ratio was calculated using a fully automated algorithm applied to time-aligned pressure traces over the wave-free period of diastole over a minimum of five beats, before adenosine administration, as previously described.¹³ Maximal coronary hyperaemia was then induced with either continuous intravenous infusion of adenosine (standard rate, 140 $\mu\text{g/kg/min}$ for 2 min) or intracoronary (IC) bolus injections (recommended dose, 100–200 μg). Fractional flow reserve was calculated as the ratio of mean distal coronary artery pressure to mean aortic pressure during stable hyperaemia (the lowest average of three consecutive beats).

For the purpose of our model, for each patient, two different views from CA were available. Each CA was labelled by expert clinicians in order to identify the stenosis which was functionally tested.

Patients gave their informed consent on admission for data collection and future publication in an anonymous fashion. This research was approved by the local institutional review board of the leading center and from ethical committees of participating centers, where required.

Data pre-processing

Coronary angiography were retrieved by full-trained cardiologists, selecting two different standard projections for each coronary stenosis. Original frame size ranged from 512×512 to 1024×1024 pixels, while frame rate ranged from 15 to 30 frames per second (fps) according to the different angiography machines and standard practice of each participating center. Original videos' duration varied from 3 to 5 s.

Thus, during the development of STARFLOW, we faced the challenge of heterogeneity in angiographic data caused by variations in acquisition protocols across different centres. To address this issue and create a standardized dataset for robust model training, we implemented several preprocessing steps.

Firstly, we adjusted the frame rate of angiograms from various rates to a consistent 15 fps and all included angiographies were resized to 256×256 pixels. Standardization is crucial for the model to accurately

interpret temporal patterns in blood flow, as inconsistent frame rates could introduce bias in the model's predictions. Normalizing the frame rate ensures that STARFLOW learns from the physiological significance of motion rather than artifacts of imaging frequency.

Additionally, we clipped the length of the angiograms to a fixed duration (4 s, 60 frames). Clipping was necessary to accommodate the fixed input size of the deep learning architecture and to remove non-essential segments of the acquisition that do not contribute to the assessment of coronary physiology. This process also reduces computational load and memory requirements, allowing for more efficient model training.

In real-world applications, these preprocessing steps can be automated and integrated into the workflow, ensuring that the model receives input data in a consistent format without manual intervention.

Furthermore, for each patient, some clinical data were also included and processed by the network. To allow the AI model to leverage effectively clinical data, it was preprocessed as follows: age and left ventricular ejection fraction (LVEF) were split into 20 intervals and encoded as a one-hot representation (i.e. it is represented as a 20-dimensional vector containing 1 for the interval it belongs to and 0 in the other positions), indication for CA [ST segment elevation myocardial infarction (STEMI), non-ST-segment elevation myocardial infarction (NSTEMI), unstable angina, stable angina, and positive stress test] were instead represented as a 5-dimension vector containing 1 if the feature was present and -1 if absent.

AI model development

We designed and developed a deep learning model, STARFLOW (whose architecture is depicted in Figure 1), which is tailored for cardiologists to assess coronary physiology non-invasively. STARFLOW is designed to predict the presence of haemodynamically significant stenoses using binary classification against standard iFR and FFR thresholds of 0.89 and 0.80, respectively. Additionally, it provides quantitative estimations of FFR and iFR values.

The architecture of STARFLOW incorporates a series of three-dimensional (3D) convolutional layers, which are digital filters that process angiographic image sequences to extract spatial and temporal features. These layers are organized following a 3D ResNet-18¹⁴ configuration, a proven framework for video analysis. The model also integrates attention mechanisms, through transformer-based modules,¹⁵ which allow the network to focus on the most relevant features across the image sequences, much like a clinician would concentrate on areas of interest.

The features extracted by the hybrid conv-transformer backbone are then fed to a three-branch layer that performs binary classification and quantification of either FFR (for the data for which FFR is estimated) or iFR value (for the data for which iFR is computed).

STARFLOW training

To validate the accuracy of STARFLOW, we employed a five-fold nested cross-validation strategy. This approach ensures that the model's performance is robust across different subsets of data. For each split, 60% of the data were used for training, 20% for validation, and the remaining 20% as a test set, taking into account the original label proportion.

The angiographic inputs were preprocessed by normalization and standardization to a uniform scale. Data augmentation techniques, including random flipping and rotation, were applied to enhance the model's ability to generalize from varying angiographic orientations.

During training, the two regression branches are not necessarily both activated, as all the samples do not have both FFR and iFR values. The specific branch is activated/trained only if the label is available for that sample. Further details about the model development and training are provided in the Supplementary methods (see [Supplementary material online, Appendix](#))

We finally assessed the contribution that clinical data (age, LVEF, chronic kidney disease, and ACS vs. CCS) could give to the obtained results. To allow STARFLOW to process this data, its architecture was slightly modified to cope with the two angiography views and the additional clinical data. In particular, clinical data were embedded, through a fully (learnable) connected layer, to a 256 feature dimension vector. Then a late fusion process between clinical and visual features is carried out before the output heads.

Study endpoints

Starting from the evaluation of sensitivity [TP/(TP + FN)] and specificity [TN/(TN + FP)], the primary endpoint was the balanced accuracy of STARFLOW, expressed as the sum of specificity and sensitivity, divided by two. We also assessed STARFLOW performance via receiver operating characteristics area under the curve (ROC-AUC) across the whole cohort of patients undergoing invasive physiological assessment with either FFR or iFR. As a secondary endpoint, we assessed the performance of the STARFLOW across the subgroup of patients investigated via FFR or iFR as separate. Moreover, also precision-recall curves were performed, showing the tradeoff between precision and recall for different threshold; a high area under that curve represents both high recall (i.e. low false negative rate) and high precision (i.e. low false positive rate). We also evaluated the performance of the model in terms of regression, i.e. the capability of the model to quantify iFR and FFR as continuous values. The adopted metrics for the regression task are mean square error (MSE) and mean absolute error (MAE).

Furthermore, we evaluated whether clinical data may lead to a better classification outcome. Accordingly, we modified STARFLOW in order to process clinical data along with the two angiography views exams.

A dedicated sensitivity analysis was performed to assess the performance of the model according to several X-ray systems of centers participating in data collection.

Finally, the performance of the present model was compared with recent state-of-the-art methods performing the same investigation with two angiography views, namely direct quantification,¹⁶ direct quantification with attention,¹⁷ and full left ventricle quantification.¹⁸

Results

Baseline features

Out of 763 patients evaluated for an intermediate coronary stenosis initially included, 389 were retained in the final dataset after a data quality screening step. During this phase, exclusion criteria were: (i) patients ($n = 106$) with only one view available (STARFLOW

Table 1 Baseline features: STEMI, ST-segment elevation myocardial infarction; NSTEMI, non ST-segment elevation myocardial infarction.

Baseline features	$n = 389$
Age, (years)	67.9 ± 9.61
Female	86 (22%)
Chronic kidney disease	66 (17%)
Left ventricular ejection fraction, %	52 ± 10
Indication for coronarography	
- STEMI	23 (5.9%)
- NSTEMI	62 (16.1%)
- Unstable angina	67 (17.2%)
- Stable angina	125 (32.1%)
- Positive stress test	112 (28.7%)
Vessel	
- Left main	13 (3.3%)
- Left anterior descending	279 (71.7%)
- Left circumflex	44 (11.2%)
- Ramus intermedius	10 (2.5%)
- Diagonal branch	8 (2.2%)
- Obtuse marginal branch	19 (4.9%)
- Postero-lateral branch	3 (0.8%)
- Right coronary artery	13 (3.4%)
Lesion location	
- Proximal	52.2%
- Mid	43.4%
- Distal	4.4%
Length of lesion, mm	18.4 ± 14.5
Diameter of stenosis, %	59.8 ± 11.1
Fractional flow reserve	0.84 ± 0.08
Instantaneous wave-free ratio	0.90 ± 0.07
Adenosine administration route*	
- Intracoronary	227 (90.4%)
- Intravenous	24 (9.6%)

* referred only to patients undergoing fractional flow reserve.

processes two views simultaneously), or (ii) patients ($n = 98$) with insufficiently accurate iFR and FFR assessments, or (iii) patients ($n = 170$) with data available only in video format (i.e. missing raw data, appendix, and web only [Figure 1](#)).

Among the 389 exams, 161 (41.4%) had only FFR values, 138 (35.5%) only iFR values, and 90 (23.1%) had both estimates.

Baseline and procedural features are summarized in [Table 1](#). Mean age was 67.9 ± 9.6 years and 86 patients (22%) were female. More than half of patients underwent CA for CCS (50.8%), while the remaining were assessed for non-culprit stenosis following ACS. The large majority (71.7%) of lesions assessed with IC physiology concerned the left anterior descending artery (LAD). Mean percentage of diameter stenosis was $59.8 \pm 11.1\%$. The histograms showing the distribution of the diameter of stenosis across the cohort are reported in appendix, web only [Figure 2](#). Mean FFR was 0.84 ± 0.08 , while mean iFR was 0.90 ± 0.07 .

Results in terms of classification accuracy by STARFLOW are reported in [Table 2](#) and in [Figure 2](#): an accuracy of 87.3% (81.2–93.4%) was detected, with a sensitivity of 82.4% (71.9–96.4%) and a specificity of 92.2% (90.4–93.9%). Receiver operating characteristics area under the curve was 0.93 for the overall cohort [95% confidence interval (CI): 0.90–0.96]. [Table 2](#) and [Figure 3](#) report the performance of STARFLOW stratified for exploited methodology: in particular, for FFR accuracy was 84.8% (77.8–91.8%), with a sensitivity of 81.9% (69.4–94.4%) and a specificity of 87.7% (93.9–96.4%) (ROC-AUC

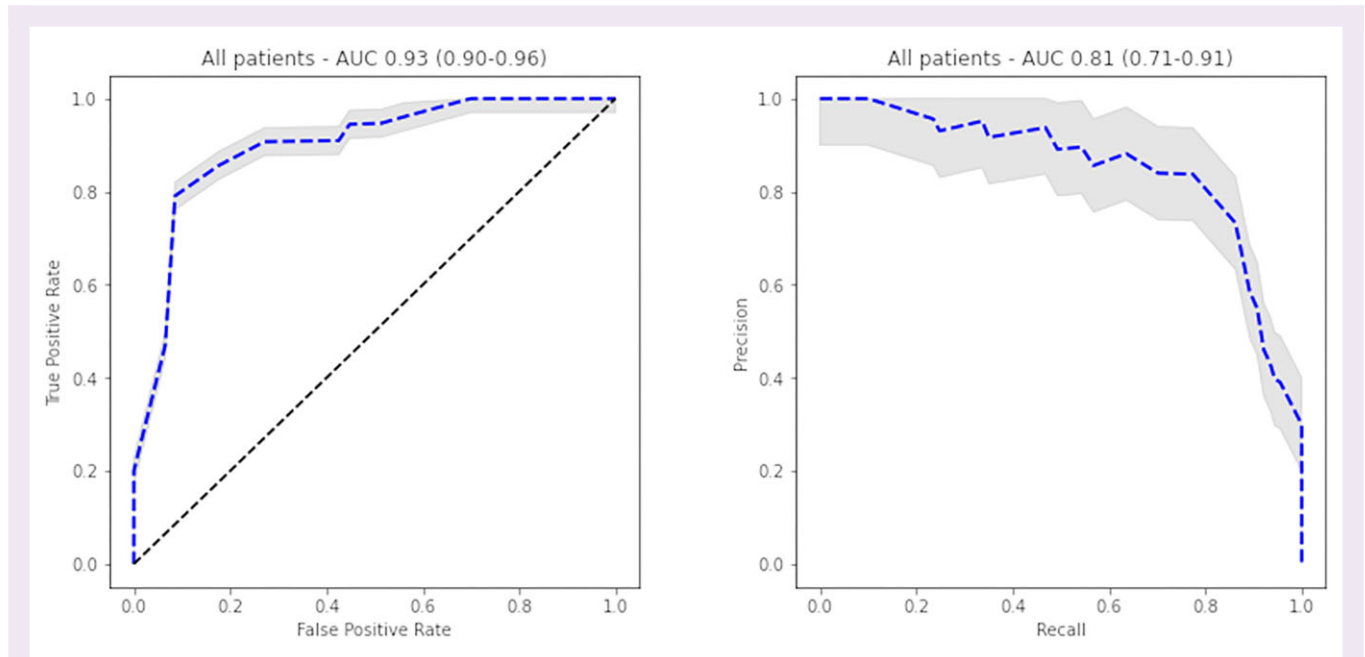


Figure 2 Area under the curve receiver operating characteristic (AUC-ROC) (left) and precision-recall curve on a fold (right) for all patients.

0.79, 95% CI 0.66–0.92). For iFR accuracy was 90.2% (86.0–94.6%), with a sensitivity of 87.2% (76.6–97.8%) and a specificity of 93.2% (91.7–94.7%, all CI 95%) (ROC-AUC for iFR 0.98, 95% CI 0.96–1.00).

We then evaluated the performance of STARFLOW when regressing continuous FFR and iFR values with results reported in [Table 3](#): STARFLOW attains an error of 0.04 for FFR and 0.02 for iFR, showing how it indeed exploits FFR and iFR values for dichotomy classification (see [Table 3](#)). STARFLOW takes approximately 62.56 ms to process a video of 60 frames, thus supporting the model's efficiency and potential for real-time application in clinical practice.

Results of the models embedding also clinical features are shown in [Table 2](#) and indicate that, in all scenarios, they partially negatively affect performance, degrading the quality of the learned visual latent space used for iFR and FFR over threshold classification. In particular, sensitivity was affected, decreasing from 82.4% (71.9–96.4) to 69.9% (56.9–82.8).

Moreover, as displayed in [Figure 4](#), [Supplementary material online, Table S1](#) and [Figure 3](#) of appendix, we compared STARFLOW with direct quantification,¹⁴ direct quantification with attention,¹⁵ and full left ventricle quantification¹⁶: STARFLOW significantly performed better than each of them.

The overall performance of STARFLOW was consistent across several X-ray acquisition systems (accuracy ranging from 87.4 ± 6.0 to 90.1 ± 5.9) (see [Supplementary material online, Table S2](#)).

Discussion

In the present study, we evaluated the feasibility and performance of an AI-based algorithm to estimate FFR/iFR values for patients with CAD and intermediate coronary stenoses, utilizing CA images as the primary modality, thereby aiming to reduce dependence on invasive pressure wire data for physiological assessment. The algorithm, named STARFLOW, was trained and tested from standard CA of 389 patients, each contributing with two angiographic views for each stenosis.

Our main results can be summarized as follows: (i) the STARFLOW system showed a very good accuracy in estimating the haemodynamic

importance of coronary stenoses, being characterized by a good sensitivity (82.4%) and a remarkable specificity (92.2%) across the whole cohort of stenoses assessed with FFR/iFR; (ii) the performance of the tool we propose was satisfying also when FFR and iFR values were regressed as continuous rather than as dichotomic variables; (iii) the addition of clinical data to the model negatively affected the overall performance, although a further increase in the specificity could be observed (96.6%).

Following the results of the FAME 2 trial,⁵ invasively measured FFR has been regarded as the clinical standard to indicate percutaneous revascularization over medical therapy in patients with stable coronary artery disease. The drawbacks associated with adenosine requirement to induce hyperaemia and consequent prolongation of procedural time and patients' discomfort, prompted the proposal and validation of non-hyperaemic indexes as iFR.⁸ Despite this methodological and practical improvement, the haemodynamic assessment of coronary stenoses has slowly penetrated common clinical practice. Reasons may be found in the need for additional equipment, the non-trivial hazard associated with passing an IC wire, and the need for a specific training and costs associated with probes. In this context, considerable research has been directed towards the development of non-invasive and user-friendly tools for physiological assessment of coronary artery stenoses. In 2013, Morris *et al.* proposed a first model of computational virtual FFR assessment.¹⁹ Despite the great value of this pioneering study, the model was greatly limited by the need of a complex pre-processing phase (requiring exportations of images obtained from rotational angiography and 3D reconstruction with an external workstation) and by the modest number of cases, in line with the proof-of-concept nature of the study.

An important step towards the non-invasive determination of FFR stemmed from the application of computational fluid dynamics (CFD) to coronary computed tomography angiography (CTA).^{20–22}

Results presented by Norgaard *et al.*, relying upon updated and refined version of the FFR-CT and emphasizing the quality of CTA acquisition, showed a very high diagnostic performance using the invasively measured FFR as gold standard.²² Despite FFR-CT has the potential of becoming a reliable gatekeeper to the pathway of

Table 2 Performance of STARFLOW for overall patients, and for FFR/iFR patients.

Methods	Accuracy balanced	AUC	PR score	Sensitivity % (Recall)	Specificity %	PPV %	NPV %
STARFLOW	87.3 (81.2–93.4)	0.93 (0.90–0.96)	0.81 (0.71–0.91)	82.4 (71.9–96.4)	92.2 (90.4–93.9)	72.8 (61.7–83.9)	95.4 (91.6–99.2)
STARFLOW FFR patient population	84.8 (77.8–91.8)	0.79 (0.66–0.92)	0.83 (0.76–0.91)	81.9 (69.4–94.4)	87.7 (85.5–89.9)	68.4 (56.0–80.8)	97.7 (95.4–100.0)
STARFLOW iFR patient population	90.2 (86.0–94.6)	0.98 (0.96–1.00)	0.77 (0.71–0.83)	87.2 (76.6–97.8)	93.2 (91.7–94.7)	78.7 (65.2–92.2)	94.3 (88.6–100.0)
STARFLOW + Clinical Data	82.9 (76.4–89.4)	0.898 (0.84–0.95)	0.74 (0.63–0.85)	69.9 (56.9–82.8)	96.6 (95.3–97.9)	68.3 (50.9–85.7)	92.1 (87.9–96.3)

FFR, fractional flow reserve; iFR, instantaneous wave-free ratio; AUC, area under the curve; PR, precision-recall; PPV, positive predictive value; NPV, negative predictive value. All CI in brackets are 95% confidence intervals. *BOLD.

performance of the STARFLOW tool referring to the entire cohort.

coronary revascularization, such technology is still limited by the processing times of CFD algorithms (varying from 10 min to several hours), the current costs and the need for a complex learning curve. These limitations will likely limit its wide diffusion in the near future. To partially overcome shortcomings of CFD-based FFR-CT, Coenen et al. first identified the potential of ML as a supportive instrument to virtually perform FFR-CT calculations with reduced delay, thereby allowing almost instantaneous and interactive physician interpretation.²³ Correlation between ML-based and CFD-based FFR was extremely good and, as expected, both outperformed anatomical CTA in predicting invasively assessed positive FFR. However, it should be noted that a substantial difference in accuracy by ML-FFR was observed when patients were grouped by their image quality (from 75% within the lowest tertile to 91% for the highest tertile of image quality). This underscores the importance of having high-quality images both for training the ML algorithm and to assess CAD on real-world clinical imaging data.

The current study used standard CA performed by several different cardiologists in various hospitals with different radiographic systems and non-standardized angiographic views since angulations were chosen by physicians for clinical purposes. This real-life scenario strongly adds to the potential diffusion of the algorithm.

Patients from the initial cohort not retained in the final group of CA were mostly excluded due to the lack of reliable invasive assessment or missing raw data. Only 13.9% of patients from a consecutive unselected population undergoing CA were excluded due to the absence of the minimal requirements for the STARFLOW system (namely two angiographic views), a limitation that can be easily overcome by simply taking into account this necessity in the future. This large applicability to all-comers compares very favourably with the finding that the application of a validated tool for virtual FFR assessment across retrospectively retrieved angiograms was feasible only in one third of cases.²⁴ The need for a strict compliance with systems' requirements, the prerequisite of 3D reconstruction of coronary anatomy, and the lack of an outcome-based validation, have all contributed to limit the clinical implementation of the non-invasive angiographic-based tools for functional evaluation of coronary stenoses that have been proposed so far.^{25–27} Among these approaches, the FFRangio software, stands out for having been validated also when performed on-site by independent operators blinded to pressure-wire derived FFR and for providing a reconstruction of the entire coronary tree with FFR values along each vessel. Of note, a wide range of diagnostic accuracy has been reported across various methods for deriving flow reserve from CA (varying from 83 to 92.7%).^{25,28}

To cope with the limitations of the models for non-invasive FFR assessment published so far, an early feasibility trial of CA-based and AI assisted prediction of FFR was recently suggested by Roguin et al.²⁹ Although the AutocathFFR software described in this manuscript showed a very good accuracy, comparable with the present STARFLOW algorithm, the former was tested in an extremely small cohort of only 31 patients. The present study involved a much larger patient cohort, used an internal validation dataset, also includes the assessment of iFR, and shows an outstanding level of accuracy also when validated against FFR and iFR as continuous values rather than just as binary outcomes.

To our knowledge, STARFLOW is the first tool for the non-invasive assessment of iFR from standard CA. We found a higher accuracy of our ML-based tool among lesion assessed with iFR as compared with FFR. It is likely that the differences between the hyperaemic and the full cardiac cycle assessment of FFR vs. the rest and diastolic-only assessment of iFR contribute to this finding. The response to hyperaemic agents varies greatly between individuals. Such issue may be even more considerable when hyperaemia is induced via IC administration of adenosine and in assessing stenoses of vessels supplying large amount of myocardium, thereby inducing less stable and

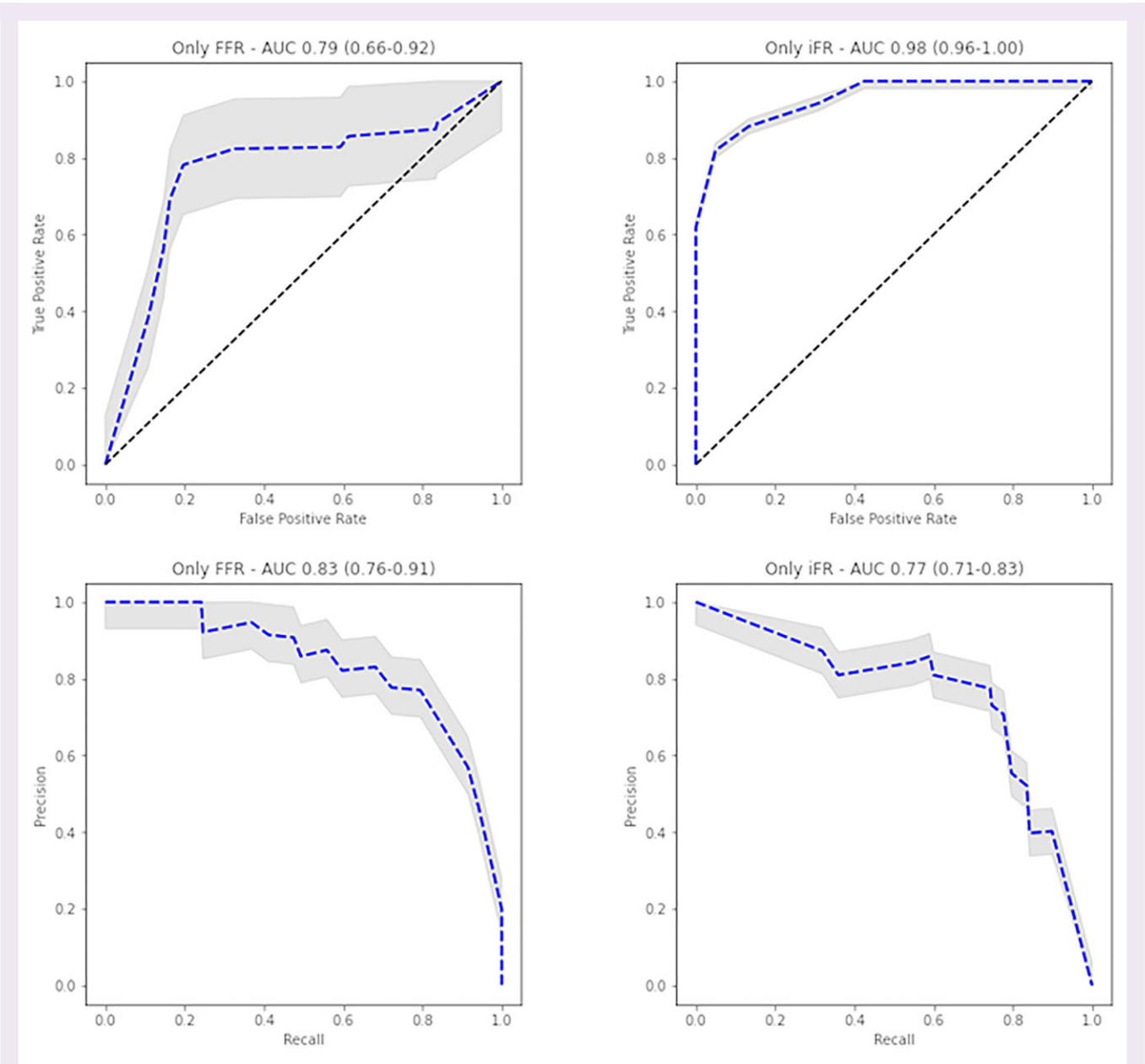


Figure 3 Area under the curve receiver operating characteristic (AUC-ROC) (above) and precision-recall curve on a fold (below) for FFR patients (left) and for iFR patients (right).

homogeneous vasodilatation. In our cohort, IC administration was indeed performed in 90% of patients undergoing FFR and most patients were assessed for stenoses of LAD. Human factors and comorbidities entailing variable responses to hyperaemia have generated difficulties in the simulation of the *in vivo* flow environment in computational models, and it may be speculated that similar problems, concerning the optimal prediction of hyperaemic indexes, may have also impaired the performance of our ML-assisted tool, resulting in a more accurate performance of the algorithm in predicting iFR.

In accordance with this hypothesis, and at variance with common sense expectations, the integration of STARFLOW with clinical data that could contribute to microvascular impairment³⁰ and therefore to the variable response to hyperaemia-inducing agents, did not result in an improved accuracy. This finding suggest that human and pharmacological factors may outweigh the clinical variables that mod-

Table 3 STARFLOW performance (in terms of MSE and MAE) when regressing FFR and iFR values.

Measure	MSE	MAE
FFR	0.060 (0.055–0.065)	0.037 (0.035–0.039)
iFR	0.045 (0.040–0.050)	0.026 (0.023–0.029)

FFR, fractional flow reserve; iFR, instantaneous waves-free ratio; MSE, mean squared error; MAE, mean absolute error.

ulates response to hyperaemia. However, such modified version of the model was characterized by a further improved specificity as compared with the main model (96.6% vs. 92.2%). As one of the

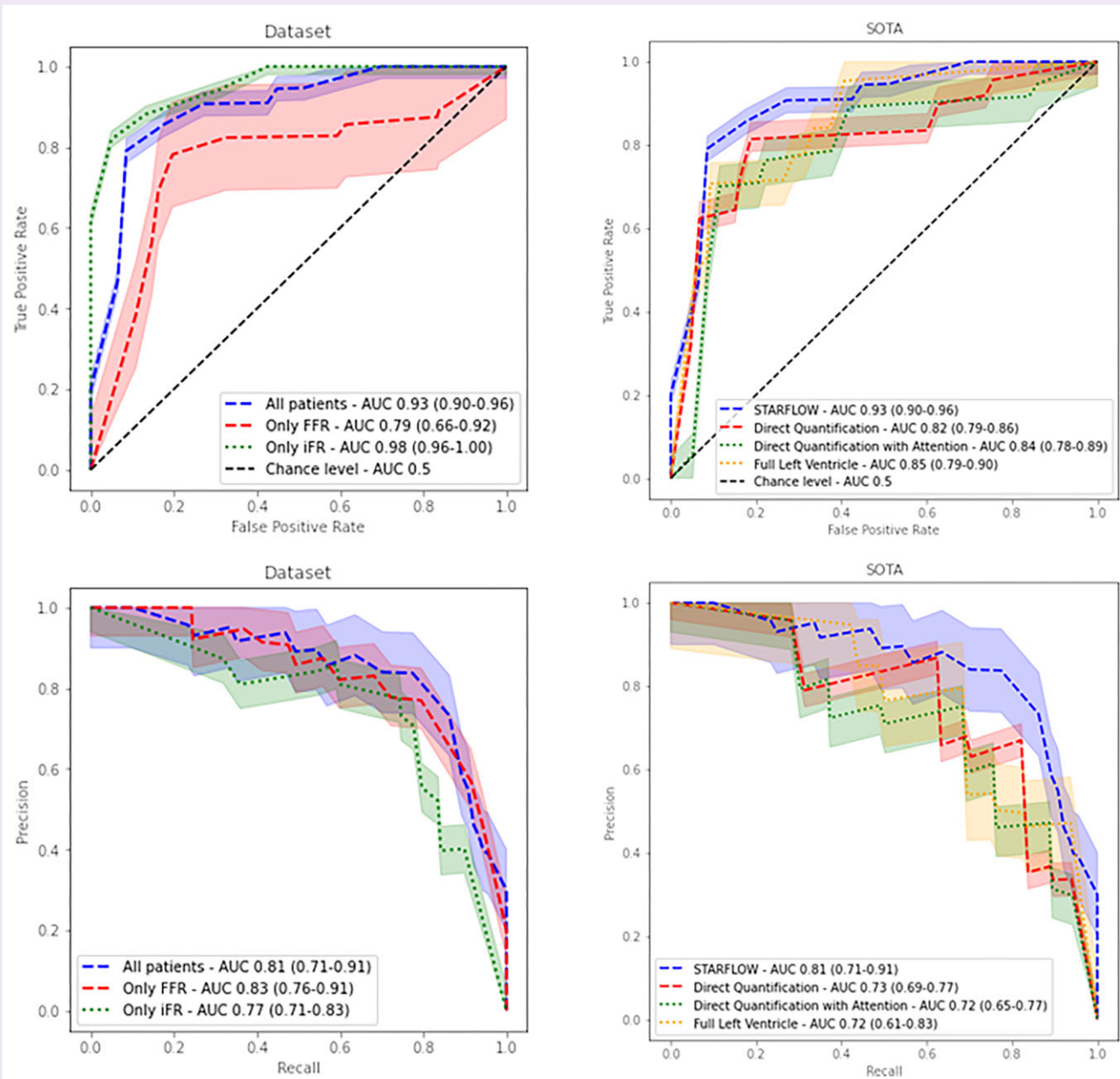


Figure 4 Area under the curve receiver operating characteristic (AUC-ROC) (above) and precision-recall curve on a fold (below) for comparing FFR/iFR datasets (left) and comparing SOTA methods (right). Central illustration: The STARFLOW model. AI, artificial intelligence; ML, machine learning; iFR, instantaneous waves-free ratio; FFR, fractional flow reserve; AUC, area under the curve; PR, precision-recall.

main limitations hampering the clinical validation of non-invasive tools, suggested so far, has been represented by the ethical risk associated with the proportion of false negatives, the tool could hopefully pursue an outcome-based validation. Indeed, STARFLOW reduces significantly the percentage (about 10% points of gain in sensitivity) of false negatives with regard to existing approaches. From a methodological standpoint, STARFLOW employs new frontiers in the AI field through a hybrid transformer-CNN architecture compared with existing solutions that instead rely only on convolutional models. This different and improved architectural design of STARFLOW allowed to obtain improved performance (especially on sensitivity) as well as better interpretability of results. Finally, it should be mentioned that as a specific advantage of such an AI-based tool, STARFLOW has the

potentiality to learn from the experience, therefore it is expected to improve its accuracy over time without being explicitly programmed.

STARFLOW in clinical practice

Despite the demonstrated superiority of revascularization guided by physiological assessment over mere angiographic estimation of coronary lesion severity, this practice remains underutilized in clinical settings. Previous research has identified barriers hindering the proper implementation of this methodology. In addition to operator reluctance and inertia in favour of more familiar practices, external barriers such as time constraints, side effects associated with drugs required for inducing hyperaemia, and the costs of pressure wire probes have

been recognized.³¹ STARFLOW, with its foundation in AI, offers a potential solution to overcome these obstacles. The system has the capacity to harness routine CA images to estimate FFR/iFR values, presenting a non-invasive alternative to traditional pressure wire measurements. The implementation of STARFLOW could be streamlined through seamless integration into existing catheterization laboratory workflows, offering clinicians an additional tool for comprehensive and cost-effective physiological evaluation. We envision STARFLOW being used in a clinical setting through a simple software interface, accessible on a standard PC. In a production scenario, the software module responsible for angiogram video analysis could be installed either locally at the hospital, on a centralized computer system with a sufficiently powerful graphics processing unit (typical prices of suitable devices range between USD 2000 and USD 4000; a single such device is enough for a hospital's needs), or remotely on a cloud infrastructure. Clinicians would access the software through a web interface with an internet browser, upload an angiogram video, and receive the coronary assessments by the deep learning model. This approach aims to be straightforward and user-friendly for clinicians, with further validation steps to ensure accuracy and reliability. Hypothetically, its integration could alleviate the need for invasive procedures and associated time constraints. While recognizing the need for further validation, we envision STARFLOW as a potential adjunct to current practices, empowering clinicians with a novel approach to physiological assessment that, upon successful validation, could contribute to broader clinical adoption.

Limitations

As most patients in the derivation dataset underwent invasive FFR/iFR for clinical reasons, a selection bias towards a relatively high burden of angiographic and functional coronary disease cannot be excluded. While the preponderance of assessments focused on the LAD artery reflects the algorithm's robust performance in a critical vascular territory, it could concomitantly highlight a limitation in terms of generalizability. Recognizing the predominance of LAD assessments, we underscore the need for caution when extrapolating the findings to vessels beyond the LAD, emphasizing the importance of further investigations encompassing a more diverse range of coronary vessels. Overall, an external validation of STARFLOW is currently missing. Although this limitation is shared by many tools for non-invasive functional assessment proposed so far, we highlight that our model was developed through a five-nested fold cross-validation where 20% of the dataset randomly served as validation set and that the reported values of sensitivity and accuracy are referred to the mean values obtained across these subsets.

Conclusions

We developed an AI-assisted based software for non-invasive assessment of FFR and iFR from standard CA. The model showed an overall very good accuracy and a remarkable specificity.

Similarly to other non-invasive tools for FFR assessment, STARFLOW may provide haemodynamic assessment of coronary stenoses without requiring the induction of hyperaemic flow, the hazard of passing an IC wire, or additional equipment, training, and costs. Differently from other tools, STARFLOW was derived across a real-life cohort of patients undergoing invasive physiological assessment without any strict clinical or angiographic exclusion criteria, is designed to require minimal additional procedure time, is capable of assessing also iFR, and does not require knowledge of vessel geometry. Albeit speculative at this stage, we hypothesize STARFLOW has the potential to improve operator and patient access to physiologically guided PCI with consequent impact on clinical outcomes and cost.

Supplementary material

Supplementary material is available at *European Heart Journal—Quality of Care and Clinical Outcomes* online.

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Conflict of interest. none declared

Data availability

The data underlying this article cannot be shared publicly as the STARFLOW software is under patent evaluation. All codes supporting STARFLOW will be privately released upon reasonable request.

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